HYPOTHESIS TESTING

Normally distributed test statistics

Testing procedures for (at least approximately ("a.l.a.")) normally distributed test statistics

- In many situations a test statistic will have a normal distribution (at least when H_0 is true, at least approximately).
- The general situation involves hypotheses about a parameter θ for which there is an estimator $\hat{\theta}$ and also $\hat{\theta} \sim N(\theta, \sigma^2_{\hat{\theta}})$ (a.l.a.).
- If θ_0 is some hypothesized value for θ then the null hypothesis is

$$H_0: \theta = \theta_0$$

some number

- If H_0 is true, then (a.l.a.)

$$Z = \frac{\hat{\theta} - \theta_0}{\sigma_{\hat{\theta}}} \sim N(0,1)$$

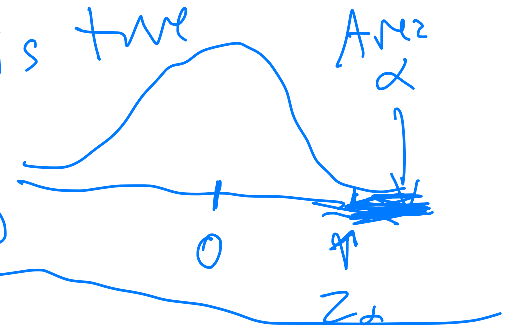
$H_0: \theta = \theta_0$

Testing procedures for (at least approximately) normally distributed test statistics

- Three possibilities for alternative hypotheses, depending on the situation:
  - $H_a: \theta > \theta_0$ ("upper tailed")

$Z = \dfrac{\hat{\theta} - \theta_0}{\sigma_{\hat{\theta}}}$ – expect this to be large if H_a is true

RR $\{Z : Z > z_\alpha\}$



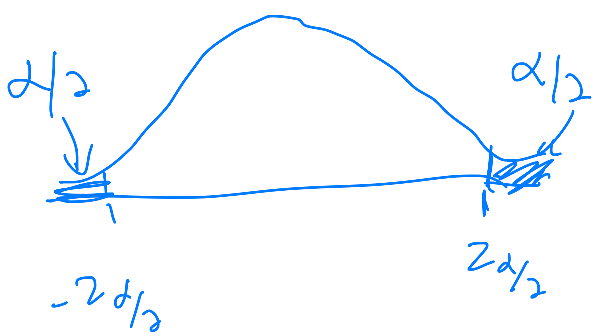
  - $H_a: \theta < \theta_0$ ("lower tailed")

RR: $\{Z : Z < -z_\alpha\}$

  - $H_a: \theta \neq \theta_0$ ("two tailed")

RR: $\{Z : Z > z_{\alpha/2}$ OR $Z < -z_{\alpha/2}\}$

$\{Z : |Z| > z_{\alpha/2}\}$



Example

$n = 25$

- Example 3: A random sample of 25 male hypertensive smokers is taken and measured for serum cholesterol level. The mean for the general population of 20- to 74-year-old males is 211 mg/100ml. If the mean of the 25 in the sample is 217, would you be justified in claiming that male hypertensive smokers have mean cholesterol levels higher than that of the larger population? (For this problem assume a normal distribution and $\sigma = 46$.)

$\sigma = 46$

μ = "true" mean of M.H.S.

$\alpha = 0.05$

$H_0: \mu = 211$

$H_a: \mu > 211$

T. S. $\bar{Y}$ - sample mean

$$\bar{Y} \sim N(\mu, 46^2)$$

Reject H_0 if $\bar{Y}$ is "large" (relative to 211)

$$Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}} = \frac{\bar{Y} - 211}{46/\sqrt{25}}$$

RR: $\{z : z > 1.645\}$ qnorm(0.95)

$$z = \frac{217 - 211}{46/\sqrt{25}} = 0.652$$

not in R.R. $\rightarrow$ fail to reject H_0

"accept H_0"

Example

- Example 4: The five-year survival rate for individuals over the age of 40 when they are diagnosed with small cell lung cancer is known to be 6.2%. To determine whether this rate is the same for patients younger than 40 when they are diagnosed, a sample of 52 patients is taken in which 6 survive at least 5 years beyond diagnosis. Can we claim that the survival rate is better than that of older patients?

p = prob. of survival (past 5 years) of <40 population

$\frac{6}{52}$

$n = 52$

$H_0: p = 0.062$

$H_a: p > 0.062$

Reject H_0 if $\hat{p}$ is "a lot bigger" than 0.062

For large n $\hat{p} \dot\sim N\left(p, \frac{p(1-p)}{n}\right)$

T.S. $\dfrac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = Z$

$\boxed{Z \dot\sim N(0,1) \text{ if } H_0 \text{ is true}}$

RR: Reject H_0 if Z "large"

$\alpha = 0.01$

$RR: \{Z : Z > z_{0.01}\} = \{Z : Z > 2.326\}$

$\hat{p} = \frac{6}{52} =$

$$Z = \frac{\frac{6}{52} - 0.062}{\sqrt{\frac{0.062(1-0.062)}{52}}} = \ldots = 1.596 \notin RR$$

$\rightarrow$ fail to reject